Module 4: Evaluation

Variable Scoping

Introducing scopes one variable at a time…

1.

Scoping 101

What's the scoop?

Bindings & Scope

Definitions

- A **binding** defines an association between a **name** and **value**.
- A **declaration** indicates that a binding exists in scope.
  - A **definition** provides the value.
- A binding's **scope** is the region where it may be accessed from.
  - Conversely, **scope** is also used to refer to the set of valid bindings.

Applications

Scopes

- Variables are the prime usage of scoping, allowing naming/reusing computed values.
- Functions also have scopes, though bindings may use more than just the name (e.g. types).
- Types also have scopes (at compile-time).
- Organizational structures like packages also involves scoping (and more, e.g. visibility).

Scope Hierarchy

Nesting

Scopes may be **nested**, forming a hierarchy of **parent** and **child** scopes.

- Examples: classes, functions, blocks (`{}`), `for` loop variables, etc.
- A child scope can access bindings declared in the parent.
  - But not the reverse!
- Variable assignment is based on the *declared* scope, not current!

Shadowing

Shadowing allows a nested child scope to override bindings in a parent scope.

- Most languages support shadowing for structural code (classes/functions).
  - Organization via namespacing.
- Languages are more selective with variables, since it can be harder to tell which is used.
  - C++ & JavaScript allow variable shadowing.
  - Java only allows variable shadowing for class fields (e.g. this.name = name).
  - Python has global/nonlocal keywords.

2.

Scope Implementation

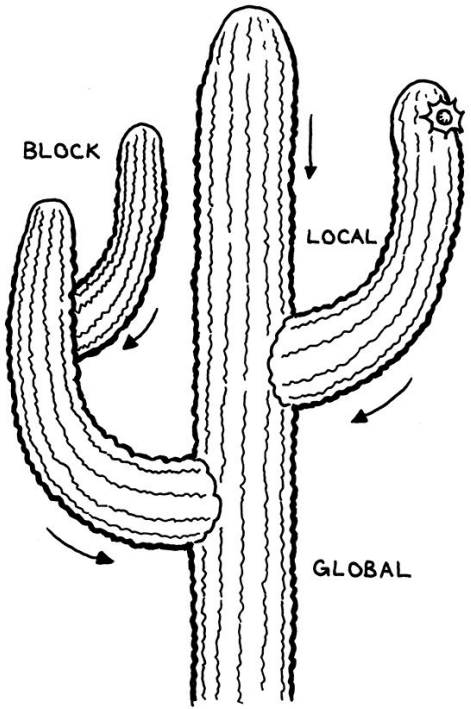
Variable access isn't magic, y'know.

“Cactus Stack” Maps

Core Idea

Use a tree of Maps to represent levels of scoping and parent/child nesting.

- Maps are great for bindings, but can’t easily support nesting.
- Include a *parent* Scope (Map) that contains higher-level bindings.
  - If not defined, recurse to parent.
  - Resolution is thus O(h).
- The result encodes a tree of scopes.

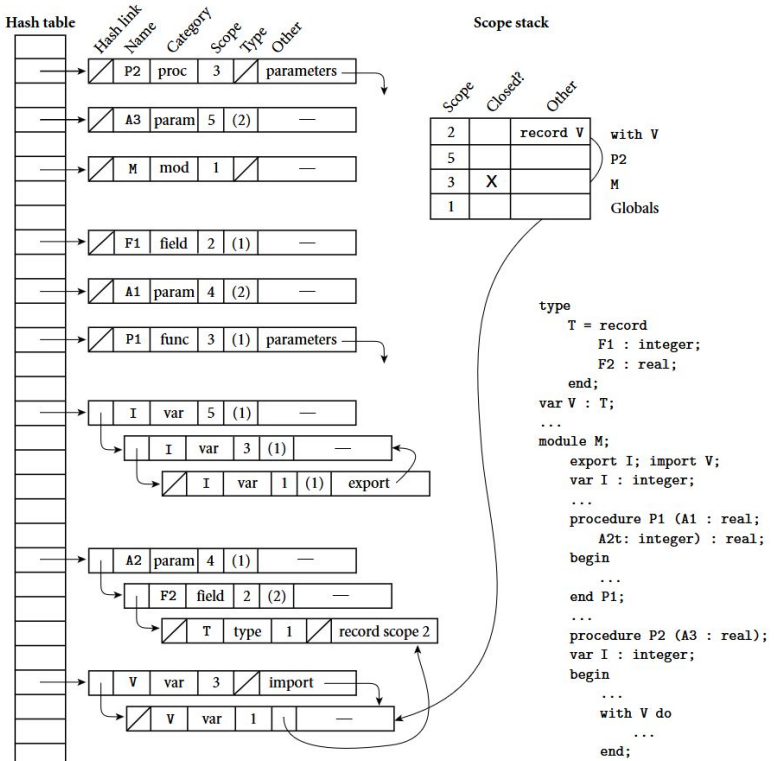


Leblanc-Cook Symbol Table

Core Idea

Use a single table for bindings and use a Scope id to prioritize nested scopes.

- A large table stores all bindings.
  - Hash table = O(1)-ish lookup.
- A Scope id is used for nesting:
  - Entering a scope adds bindings to the start of the chain (linked list).
    - Resolution picks the first value.
  - Exiting a scope is more expensive as bindings must be removed.



Compiled languages can do better.

Compilers will allocate memory on the stack/heap.

Variables can be renamed to “flatten” scopes.

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